

Enhancing the Automotive Research Experience: An Automotive Brand Summarizer Utilizing Retrieval-Augmented Generation

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Problem Overview

The automotive industry can be daunting to navigate for inexperienced shoppers. The variety of makes, models, and trim levels can present shoppers with an overwhelming amount of information. Even when knowledgeable in this information, shoppers may still have lingering questions on the quality of automobile between brands, as some are considered more reputable than others, as a result of either historical sentiment or verified data such as problems per 100 vehicles (PP100). As such, the process of navigating the car-buying experience is a holistic one, taking into account features, performance, and reliability, amongst other criteria. The aim of this experiment is to create an interactive model, powered by Retrieval-Augmented Generation (RAG), to quickly deliver prospective buyers a summary of brand features and reliability, using both general brand sentiment and reviews of specific models of vehicles. The model would use user-input in the form of a question as the query for the model's answer.

Data

The data used for this experiment consists of general brand overviews, general brand reviews, summary rankings and their explanations, and a variety of reviews from reputable automotive journalist publications. Sources include US News, Motor Trend, Consumer Reports, J.D. Power, Kelley Blue Book, and Edmunds. Data was collected via the respective source's webpages and organized into an input CSV file, with labels for each associated brand given an entry. 100 entries from various sources were used for model development, each corresponding to one of ten brands: Honda, Toyota, Subaru, Jeep, B.M.W., Mazda, Tesla, Dodge, Ford, and Chevrolet.

Approach

The method employed in this experiment was a Retrieval-Augmented Generation (RAG) model, with the input data ultimately being used as the corpus from which answers would be based on. This method was ultimately chosen over fine-tuning a model due to data quantity limitations. Language model selection was limited to smaller models from the HuggingFace transformers library, due to compute resource limitations. The same selection process was applied when choosing the embedding model. For initial implementation, the model-embedder pair chosen as a baseline was Google's T5-Small transformer, combined with the all-MiniLM-L6-V2 embedder. This would serve as a reference point for evaluation and model selection. Other language models to be considered were Google's T5-Base and Facebook's bart-large-cnn. Likewise with embedders, competing models were multi-qa-MiniLM-cos-v1 and all-mpnet-base-v2.

Results being evaluated include ROUGE scores (ROUGE-1, ROUGE-2, and ROUGE-L) and BLEU scores. Evaluations were generated for each model-embedder combination against a set of test queries and their reference answers. These results are discussed in the next section.

The data preprocessing pipeline consisted of first taking the CSV file and breaking the text into chunks of 50 tokens, allowing for 5 tokens of overlap between each chunk. The brand associated to each chunk was retained to maintain context even in the absence of explicit reference. This allowed for the original 100 documents to be expanded into 309 entries for retrieval. The data was then passed to a tagging function, where a list of consumer topics such as reliability, performance, and comfort was defined with associated keywords. Documents containing these keywords would be tagged with a given category to improve document relevance. Embeddings were generated using the selected embedder and Facebook AI Similarity Search (FAISS) library: chunks were vectorized and then normalized, which allowed for indices to be generated using FAISS IndexFlatIP, which computes cosine similarity. These preprocessing steps combined allowed for a robust method of finding relevant documents to be gathered upon request.

The user query processing pipeline took a query and extracted the relevant brand through comparison with a predefined list of brands from the input data. In a similar fashion, the topic of the query was extracted to narrow the document search to any relevant tags. Following brand and topic extraction, the top 7 most relevant documents from the corpus were selected for answer generation and passed to the associated functions.

Results and Demonstration

Testing each model-embedder pair produced many results for individual test queries. ROUGE and BLEU scoring functions were implemented to evaluate and log performance data for each test case. The overall dataset is shown in Figure 1:

	model	embedder	question	rouge-1	rouge-2	rouge-L	bleu
0	t5-small	all-MiniLM-L6-v2	Is Toyota reliable?	0.255319	0.000000	0.170213	0.010774
1	t5-small	all-MiniLM-L6-v2	Does BMW make good performance cars?	0.119403	0.000000	0.119403	0.010796
2	t5-small	all-MiniLM-L6-v2	Are Honda cars fuel efficient?	0.326531	0.042553	0.204082	0.027336
3	t5-small	multi-qa-MiniLM-L6-cos-v1	Is Toyota reliable?	0.255319	0.000000	0.170213	0.010774
4	t5-small	multi-qa-MiniLM-L6-cos-v1	Does BMW make good performance cars?	0.125000	0.000000	0.093750	0.011943
5	t5-small	multi-qa-MiniLM-L6-cos-v1	Are Honda cars fuel efficient?	0.297872	0.044444	0.170213	0.024061
6	t5-small	all-mpnet-base-v2	Is Toyota reliable?	0.255319	0.000000	0.170213	0.010774
7	t5-small	all-mpnet-base-v2	Does BMW make good performance cars?	0.121951	0.000000	0.121951	0.008503
8	t5-small	all-mpnet-base-v2	Are Honda cars fuel efficient?	0.216216	0.114286	0.216216	0.014247
9	t5-base	all-MiniLM-L6-v2	Is Toyota reliable?	0.192308	0.000000	0.115385	0.009728
10	t5-base	all-MiniLM-L6-v2	Does BMW make good performance cars?	0.144928	0.000000	0.144928	0.011197
11	t5-base	all-MiniLM-L6-v2	Are Honda cars fuel efficient?	0.600000	0.583333	0.600000	0.427765
12	t5-base	multi-qa-MiniLM-L6-cos-v1	Is Toyota reliable?	0.192308	0.000000	0.115385	0.009728
13	t5-base	multi-qa-MiniLM-L6-cos-v1	Does BMW make good performance cars?	0.123457	0.000000	0.098765	0.009249
14	t5-base	multi-qa-MiniLM-L6-cos-v1	Are Honda cars fuel efficient?	0.081633	0.000000	0.081633	0.009724
15	t5-base	all-mpnet-base-v2	Is Toyota reliable?	0.192308	0.000000	0.115385	0.009728
16	t5-base	all-mpnet-base-v2	Does BMW make good performance cars?	0.151515	0.000000	0.090909	0.011420
17	t5-base	all-mpnet-base-v2	Are Honda cars fuel efficient?	0.181818	0.095238	0.181818	0.012146
18	facebook/bart-large-cnn	all-MiniLM-L6-v2	Is Toyota reliable?	0.315789	0.000000	0.210526	0.013442
19	facebook/bart-large-cnn	all-MiniLM-L6-v2	Does BMW make good performance cars?	0.195122	0.000000	0.121951	0.009411
20	facebook/bart-large-cnn	all-MiniLM-L6-v2	Are Honda cars fuel efficient?	0.169492	0.000000	0.101695	0.009909
21	facebook/bart-large-cnn	multi-qa-MiniLM-L6-cos-v1	Is Toyota reliable?	0.315789	0.000000	0.210526	0.013442
22	facebook/bart-large-cnn	multi-qa-MiniLM-L6-cos-v1	Does BMW make good performance cars?	0.206897	0.000000	0.160920	0.009094
23	facebook/bart-large-cnn	multi-qa-MiniLM-L6-cos-v1	Are Honda cars fuel efficient?	0.226415	0.039216	0.150943	0.019845
24	facebook/bart-large-cnn	all-mpnet-base-v2	Is Toyota reliable?	0.315789	0.000000	0.210526	0.013442
25	facebook/bart-large-cnn	all-mpnet-base-v2	Does BMW make good performance cars?	0.111111	0.000000	0.055556	0.009909
26	facebook/bart-large-cnn	all-mpnet-base-v2	Are Honda cars fuel efficient?	0.196078	0.081633	0.196078	0.019407

Figure 1: ROUGE and BLEU scores for Model-Embedder Combinations.

Given no clear winner for a combination, average scores were calculated and visualized, as shown in Figures 2-10:

	model	rouge-1	rouge-2	rouge-L	bleu
0	facebook/bart-large-cnn	0.228054	0.013428	0.157636	0.013100
1	t5-base	0.206697	0.075397	0.171579	0.056743
2	t5-small	0.219215	0.022365	0.159584	0.014356

Figure 2: Average ROUGE Scores by Model.

	embedder	rouge-1	rouge-2	rouge-L	bleu
0	all-MiniLM-L6-v2	0.257655	0.069543	0.198687	0.058929
1	all-mpnet-base-v2	0.193567	0.032351	0.150961	0.012175
2	multi-qa-MiniLM-L6-cos-v1	0.202743	0.009296	0.139150	0.013096

Figure 3: Average BLEU Score by Model.

	model	embedder	rouge-2
0	t5-base	all-MiniLM-L6-v2	0.194444
1	t5-small	all-mpnet-base-v2	0.038095
2	t5-base	all-mpnet-base-v2	0.031746
3	facebook/bart-large-cnn	all-mpnet-base-v2	0.027211
4	t5-small	multi-qa-MiniLM-L6-cos-v1	0.014815
5	t5-small	all-MiniLM-L6-v2	0.014184
6	facebook/bart-large-cnn	multi-qa-MiniLM-L6-cos-v1	0.013072
7	facebook/bart-large-cnn	all-MiniLM-L6-v2	0.000000
8	t5-base	multi-qa-MiniLM-L6-cos-v1	0.000000

Figure 4: Average ROUGE-2 Score by Model-Embedder Pair, Sorted.

	model	embedder	rouge-L
0	t5-base	all-MiniLM-L6-v2	0.286771
1	facebook/bart-large-cnn	multi-qa-MiniLM-L6-cos-v1	0.174130
2	t5-small	all-mpnet-base-v2	0.169460
3	t5-small	all-MiniLM-L6-v2	0.164566
4	facebook/bart-large-cnn	all-mpnet-base-v2	0.154053
5	t5-small	multi-qa-MiniLM-L6-cos-v1	0.144725
6	facebook/bart-large-cnn	all-MiniLM-L6-v2	0.144724
7	t5-base	all-mpnet-base-v2	0.129371
8	t5-base	multi-qa-MiniLM-L6-cos-v1	0.098594

Figure 5: Average ROUGE-L Score by Model-Embedder Pair, Sorted.

	model	embedder	bleu
0	t5-base	all-MiniLM-L6-v2	0.149564
1	t5-small	all-MiniLM-L6-v2	0.016302
2	t5-small	multi-qa-MiniLM-L6-cos-v1	0.015593
3	facebook/bart-large-cnn	all-mpnet-base-v2	0.014253
4	facebook/bart-large-cnn	multi-qa-MiniLM-L6-cos-v1	0.014127
5	t5-small	all-mpnet-base-v2	0.011175
6	t5-base	all-mpnet-base-v2	0.011098
7	facebook/bart-large-cnn	all-MiniLM-L6-v2	0.010920
8	t5-base	multi-qa-MiniLM-L6-cos-v1	0.009567

Figure 6: Average BLEU Score by Model-Embedder Pair, Sorted.

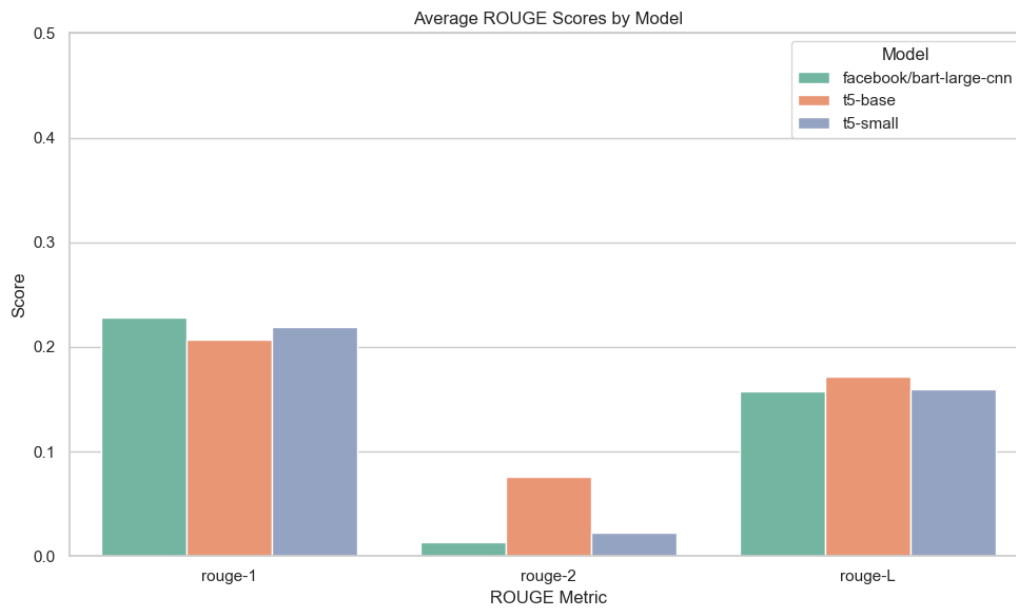


Figure 7: Average ROUGE Score by Model.

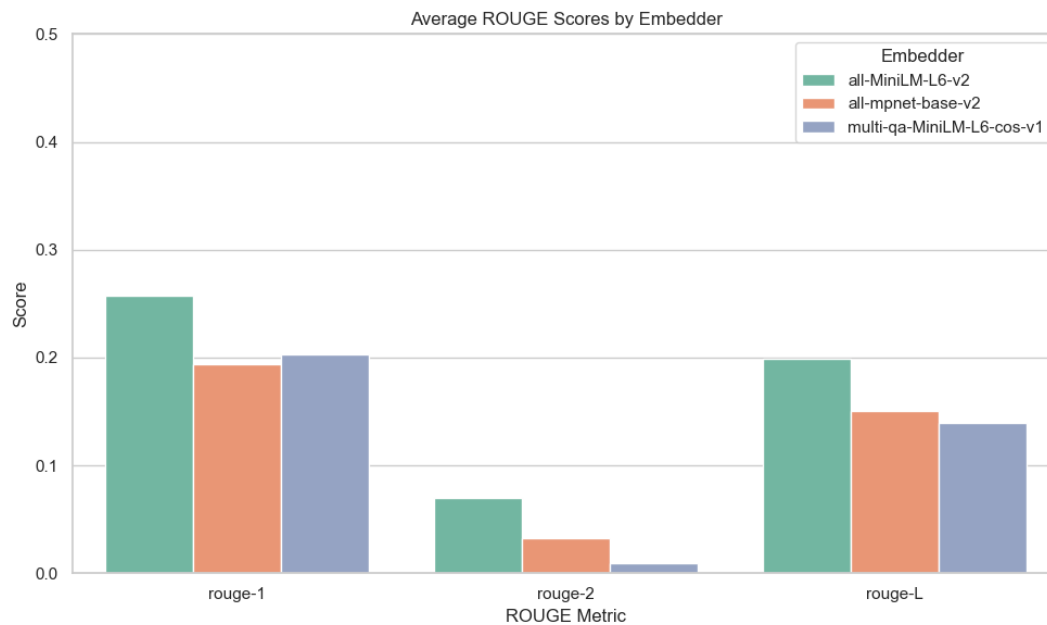


Figure 8: Average ROUGE Score by Embedder.

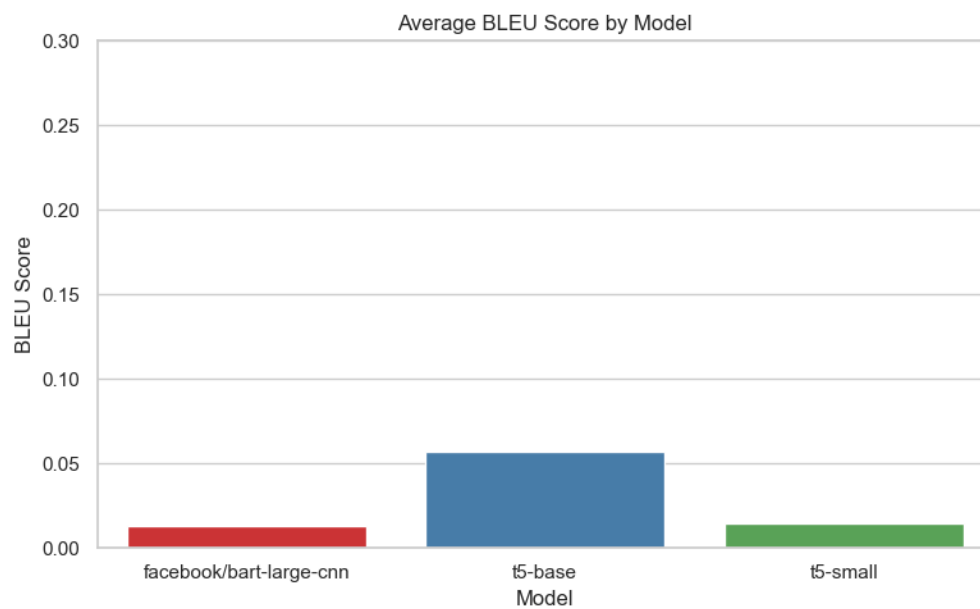


Figure 9: Average BLEU Score by Model.

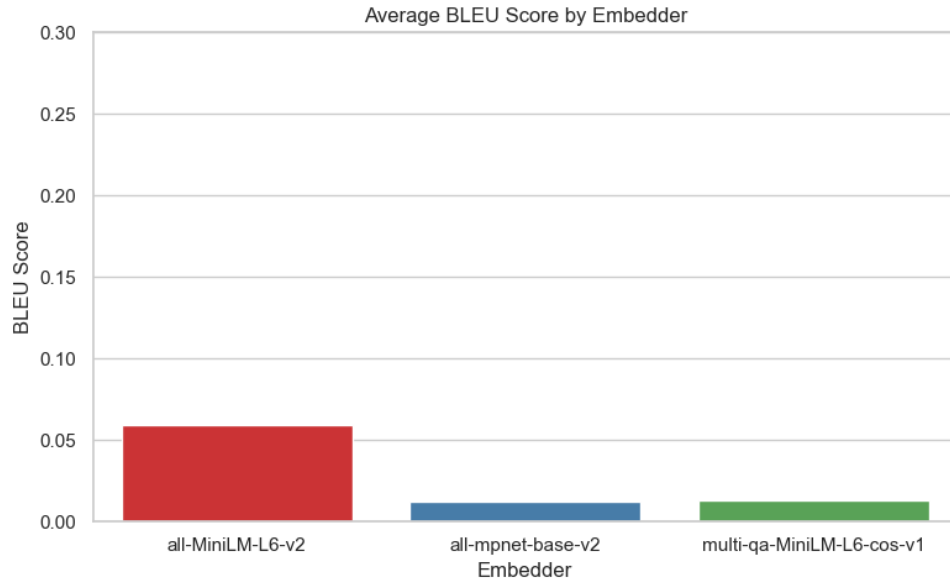


Figure 10: Average BLEU Score by Embedder.

Ultimately, the model-embedder pair chosen for the final RAG system was the T5-Base model with the all-MiniLM-L6-v2 embedder, showing advantages in handling longer strings of text. This also aligned with the subjective evaluations of model output, evaluating the overall grammatical correctness and cohesiveness of the generated responses.

While not perfect, the responses generated by the model show promise. A sample demonstration of the model via a Flask application is shown in figures 11-13:

← → ↺ 127.0.0.1:5000 ☆

Auto Reviewer

Ask about a car brand (e.g., 'Is Honda reliable?')

Available brands to ask about: Honda, Toyota, Subaru, Jeep, BMW, Mazda, Tesla, Dodge, Ford, Chevrolet

Automotive Reviewer

Question:

Answer = many Honda: Hondas are generally affordable, reliable, and fuel-efficient, with intuitive controls and good driving characteristics . its SUVs and sedans are usually near the top of Consumer Reports' ratings for road tests and predicted reliability .

Figure 11: Sample Response from Model. Query: “Is Honda reliable?”

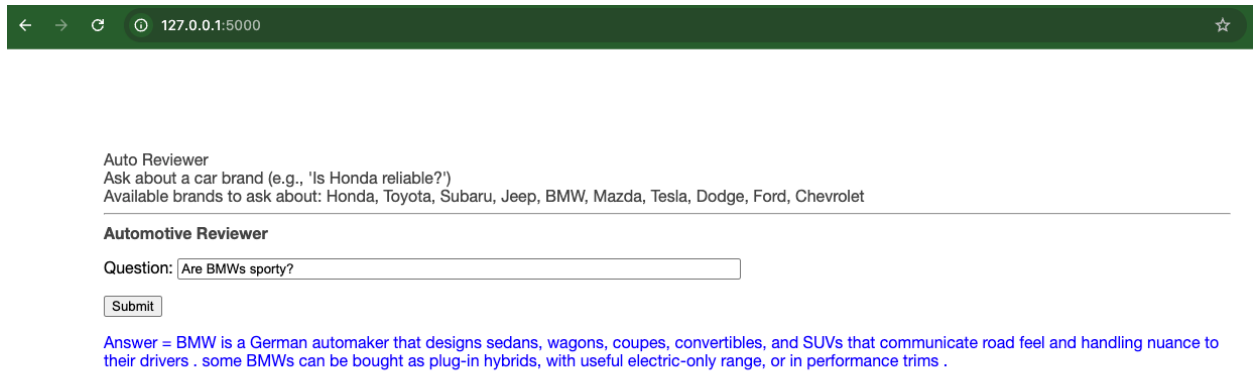


Figure 12: Sample Response from Model. Query: “Are BMWs sporty?”

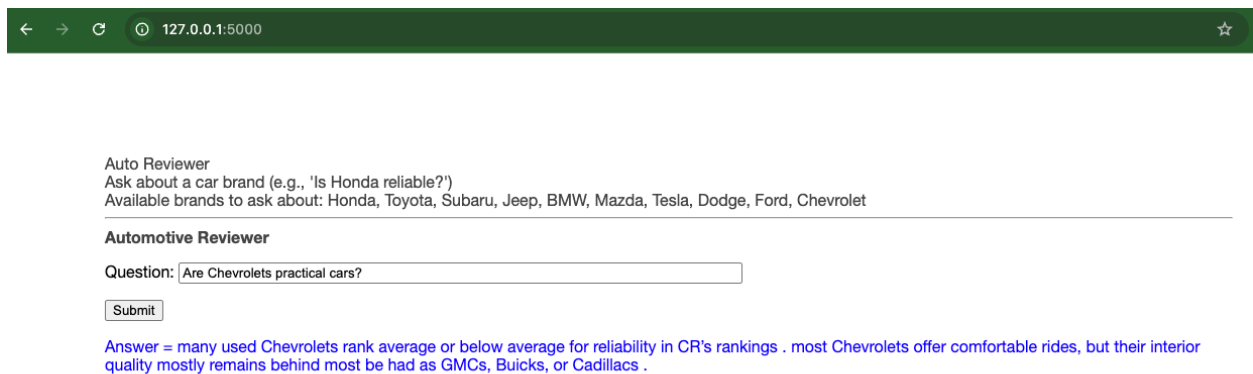


Figure 13: Sample Response from Model. Query: “Are Chevrolets practical cars?”

Related Work

S. E. Shukri, R. I. Yaghi, I. Aljarah and H. Alsawalqah (2015). ”*Twitter sentiment analysis: A case study in the automotive industry*”: This article looks at using Twitter data to understand customer sentiment analysis between car manufacturers. The aim of this experiment is to implement a Naïve-Bayes classification model to classify Twitter data as positive or negative for a selected number of brands, presenting the percent of positive tweets as a result statistic [1]. This varies from my experiment in that it aims to present a binary classification, rather than a holistic RAG model.

Dasgupta, P., Banerjee, A., Gusain, R., Saxena, U., & Jain, K. K. (2022). ” *Sentiment Analysis for Car Buying in a Post-COVID World.*”: The aim of this experiment was to track the changes in sentiment towards the car-buying process following tapering COVID-19 restrictions. N-Gram

analysis was applied to social media data to collect customer sentiment during and following COVID-19 restrictions [2]. My project differs in that I do not plan to evaluate sentiment as a function of time but instead use recent sentiment as a focus in data collected.

Alexey Uglov, Felician Campean, Amr Abdullatif, Daniel Neagu, Aleksandr Doikin, David Delaux, Pascal Bonnaud (2024). *“An NLP-based framework for early identification of design reliability issues from heterogeneous automotive lifecycle data.”*: The aim of this experiment was to apply Natural Language Processing (NLP) to determine common causes of warranty claims for a variety of automotive brands. This experiment utilized Naïve-Bayes and Xgboost approaches to build the associated model [3]. This differs from my experiment in that the focus of this study is to improve service-side identification of common issues for various vehicles, rather than general consumer/journalist sentiment.

D. Vicente-Estévez, A. Dacal-Nieto, A. Paradela, V. Alonso-Ramos, J. J. Areal and A. Riomaio (2024). *“Categorisation of Post-Sale Automotive Customer Feedback Using Natural Language Processing.”*: This study aims to categorize customer feedback following a new car sale to find common issues and improve the manufacturing process [4]. This is similar to my experiment in that it aims to identify points of criticism, but from a manufacturing/service perspective rather than a general quality perspective.

Zhang, Y., Larsson, A., Larsson, T., Tian, W., Zhang, L., Wang, W. (2024). *“Utilizing Natural Language Processing for Enhancing Collaborative Value-Driven Design of Smart Product Service System: Smart E-Vehicle Application.”*: This article aims to use NLP as a means of summarizing customer feedback and desired use-cases to better align vehicle design. The model designed for this was a combination of BERT and VDD models [5]. This differs from my experiment in that it is directly evaluating customer feedback to assist manufacturers cater to the market, rather than summarizing sentiment for the direct benefit of the consumer.

Conclusions and Future Considerations

The overall problem of guiding consumers in their search for a new car was addressed in this model. While summaries were not at a conversational level, the generated responses were able to stay on topic as it pertained to the queried brand and associated features. The pivot from fine-tuning a large language model to retrieval-augmented generation allowed for better use of an otherwise limiting dataset. Future iterations of this project may include expanding and labeling the dataset to be used in LLM fine-tuning, employing more powerful models using GPU-

accelerated compute, or expanding data gathering beyond journalistic sources and including consumer feedback. Ultimately this model serves as a first step towards a powerful tool to make a traditionally overwhelming process more comfortable from start to finish.

Works Cited

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